MSG No. 140.548 $I4'/m'cm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `4'2'2`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[5,6,7,8,13,14,15,16]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[17,18,19,20,25,26,27,28]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[21,22,23,24,29,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `-42'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,11,12,21,22,31,32]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[3,4,9,10,23,24,29,30]`
3	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[5,6,15,16,17,18,27,28]`
4	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[7,8,13,14,19,20,25,26]`

Table 3: Wyckoff site: `4c`, site symmetry: `4'/m'..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,5,6,9,10,13,14]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[3,4,7,8,11,12,15,16]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[17,18,21,22,25,26,29,30]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[19,20,23,24,27,28,31,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `m'.m'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,13,14,19,20,31,32]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[3,4,15,16,17,18,29,30]`
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[5,6,9,10,23,24,27,28]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7,8,11,12,21,22,25,26]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2/m'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,3,29,31]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,4,30,32]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,8,25,28]`
4	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[6,7,26,27]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[9,12,21,24]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[10,11,22,23]`
7	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[13,15,17,19]`

8	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[14,16,18,20]`

Table 6: Wyckoff site: `8f`, site symmetry: `4'..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,9,10]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[3,4,11,12]`
3	$[0,\ 0,\ -z]$	`[5,6,13,14]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[7,8,15,16]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[17,18,25,26]`
6	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[19,20,27,28]`
7	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[21,22,29,30]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[23,24,31,32]`

Table 7: Wyckoff site: `8g`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2,31,32]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[3,4,29,30]`
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[5,6,27,28]`
4	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8,25,26]`
5	$\left[\frac{1}{2},\ 0,\ z\right]$	`[9,10,23,24]`
6	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,12,21,22]`
7	$\left[0,\ \frac{1}{2},\ -z\right]$	`[13,14,19,20]`
8	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[15,16,17,18]`

Table 8: Wyckoff site: `8h`, site symmetry: `m'.2m'`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[1,14,19,32]`
2	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[2,13,20,31]`
3	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[3,16,17,30]`
4	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}\right]$	`[4,15,18,29]`
5	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[5,10,24,27]`
6	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[6,9,23,28]`
7	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[7,12,22,25]`
8	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[8,11,21,26]`

Table 9: Wyckoff site: `16i`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	`[1,3]`
2	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[2,4]`
3	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[5,8]`

continued ...

Table 9

No.	position	mapping
4	$\left[-x,\ x,\ \frac{3}{4}\right]$	[6,7]
5	$\left[-x,\ x,\ \frac{1}{4}\right]$	[9,12]
6	$\left[x,\ -x,\ \frac{1}{4}\right]$	[10,11]
7	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[13,15]
8	$\left[x,\ x,\ \frac{3}{4}\right]$	[14,16]
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[17,19]
10	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[18,20]
11	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[21,24]
12	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[22,23]
13	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[25,28]
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[26,27]
15	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[29,31]
16	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[30,32]

Table 10: Wyckoff site: 16j, site symmetry: .2'.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	[1,11]
2	$\left[-x,\ 0,\ \frac{1}{4}\right]$	[2,12]
3	$\left[0,\ x,\ \frac{1}{4}\right]$	[3,9]
4	$\left[0,\ -x,\ \frac{1}{4}\right]$	[4,10]
5	$\left[0,\ -x,\ \frac{3}{4}\right]$	[5,15]
6	$\left[0,\ x,\ \frac{3}{4}\right]$	[6,16]
7	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[7,13]
8	$\left[x,\ 0,\ \frac{3}{4}\right]$	[8,14]
9	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[17,27]
10	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[18,28]
11	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[19,25]
12	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[20,26]
13	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[21,31]
14	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[22,32]
15	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[23,29]
16	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[24,30]

Table 11: Wyckoff site: 16k, site symmetry: m'..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,14]
2	$[-x,\ -y,\ 0]$	[2,13]
3	$\left[y,\ x,\ \frac{1}{2}\right]$	[3,16]
4	$\left[-y,\ -x,\ \frac{1}{2}\right]$	[4,15]
5	$[y,\ -x,\ 0]$	[5,10]
6	$[-y,\ x,\ 0]$	[6,9]
7	$\left[-x,\ y,\ \frac{1}{2}\right]$	[7,12]

continued ...

Table 11

No.	position	mapping
8	$\left[x,\ -y,\ \frac{1}{2}\right]$	[8,11]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[17,30]
10	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[18,29]
11	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[19,32]
12	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ 0\right]$	[20,31]
13	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[21,26]
14	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[22,25]
15	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	[23,28]
16	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[24,27]

Table 12: Wyckoff site: 16l, site symmetry: ..m'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	[1,32]
2	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	[2,31]
3	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-z\right]$	[3,30]
4	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	[4,29]
5	$\left[x+\frac{1}{2},\ -x,\ -z\right]$	[5,27]
6	$\left[\frac{1}{2}-x,\ x,\ -z\right]$	[6,28]
7	$\left[-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[7,25]
8	$\left[x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[8,26]
9	$\left[\frac{1}{2}-x,\ x,\ z\right]$	[9,23]
10	$\left[x+\frac{1}{2},\ -x,\ z\right]$	[10,24]
11	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[11,21]
12	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[12,22]
13	$\left[-x,\ \frac{1}{2}-x,\ -z\right]$	[13,20]
14	$\left[x,\ x+\frac{1}{2},\ -z\right]$	[14,19]
15	$\left[\frac{1}{2}-x,\ -x,\ z+\frac{1}{2}\right]$	[15,18]
16	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[16,17]

Table 13: Wyckoff site: 32m, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-x,\ -y,\ z]$	[2]
3	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[3]
4	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[4]
5	$[y,\ -x,\ -z]$	[5]
6	$[-y,\ x,\ -z]$	[6]
7	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[7]
8	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	[8]
9	$[-y,\ x,\ z]$	[9]
10	$[y,\ -x,\ z]$	[10]
11	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[11]

continued ...

Table 13

No.	position	mapping
12	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	[12]
13	$[-x,\, -y,\, -z]$	[13]
14	$[x,\, y,\, -z]$	[14]
15	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[16]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[18]
19	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[19]
20	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[20]
21	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[21]
22	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[22]
23	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[23]
24	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[24]
25	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[25]
26	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[26]
27	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[27]
28	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[28]
29	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[29]
30	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[30]
31	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[31]
32	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[32]